

- 8 Osteogenesis imperfecta (OI) is a group of rare genetic disorders which results in bones that are not formed properly and can break easily.

There are several types of OI, as a number of different genes can be affected.

Some of the symptoms of OI can be life-threatening.

The commonest types of OI are caused by mutations in genes coding for the structure of collagen.

- (a) In some types of OI, the collagen is poorly formed and lacks strength.

Explain how a substitution mutation could result in collagen that lacks strength.

(4)

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- (b) In most types of OI the allele responsible for causing the condition is **dominant**.

Determine the probability that a child could inherit OI if one parent has OI and is heterozygous and the other parent does not have OI and is therefore homozygous recessive.

Use a genetic diagram to support your answer, indicating the phenotypes of the offspring and their corresponding genotypes.

(4)

Genotype of parent with OI:

Genotype of parent without OI:

Probability that a child could inherit OI

*(c) Genetic screening can be used by couples who are at risk of having a child with OI.

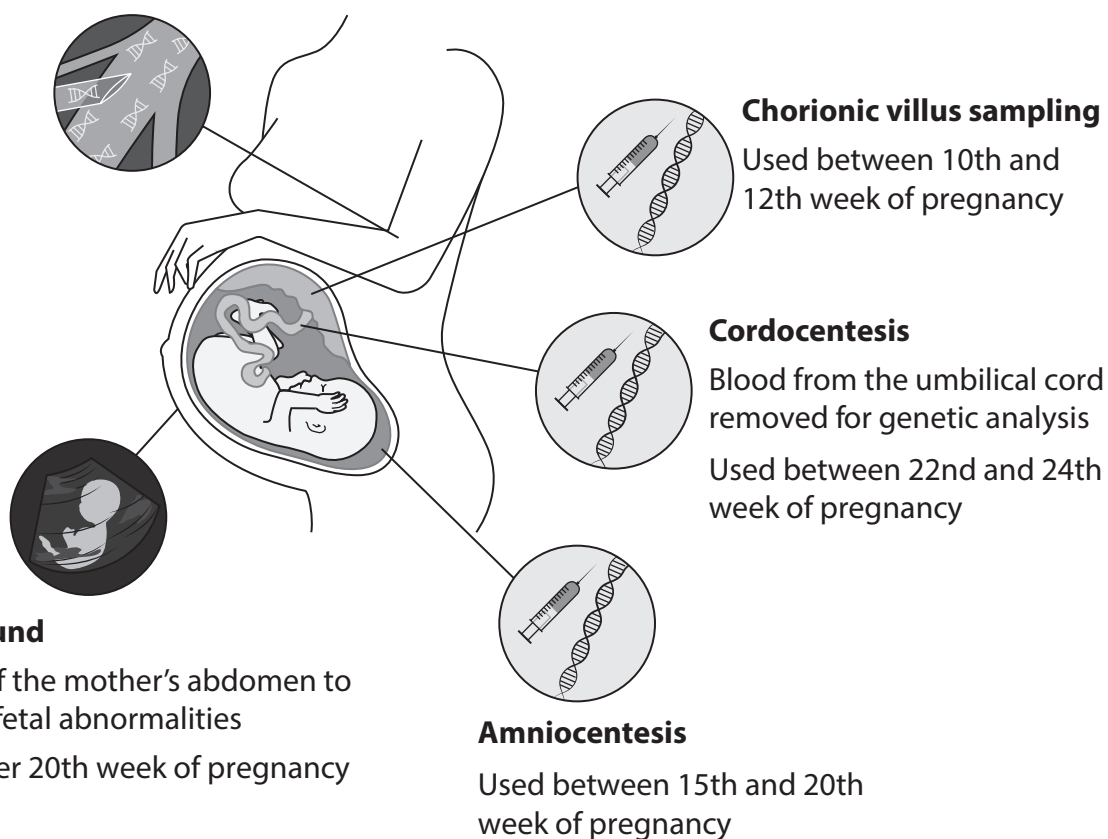
The genetic screening options include pre-implantation genetic diagnosis (PGD) and prenatal testing.

The diagram gives some information about some of the types of prenatal testing available to these couples.

NIPT

Sample of mother's blood removed and fetal DNA used for genetic analysis

Used between 7th and 10th week of pregnancy



Discuss the advantages and disadvantages of the genetic screening options available to couples at risk of having a child with OI.

Use all the information in the question and your own knowledge to support your answer.

(6)

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(Total for Question 8 = 14 marks)

TOTAL FOR PAPER = 80 MARKS

